

LAB: 08

This lab focuses on email analysis.

Objective:

- Learn E-mail investigation.

Learning Activities:

At the end of these activities, you should understand:

- How to apply digital forensics methods to investigating email communications.

Tools to use:

- Outlook client
- Aid4Mail

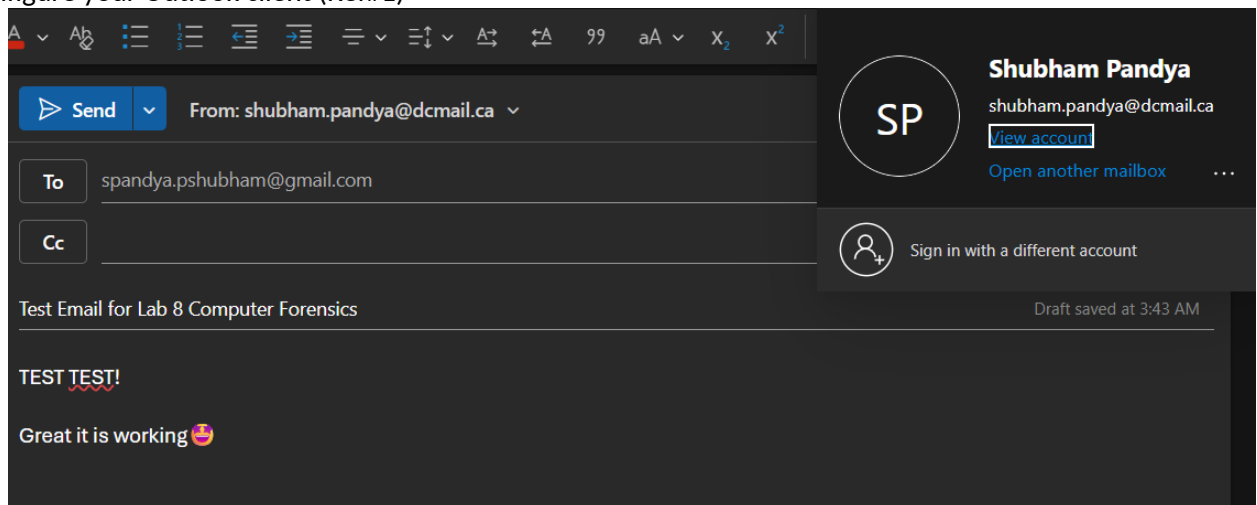
Note: Please include a screenshot of each step. Crop your screenshot to show relevant information only. Uncropped screenshots will result in a 10% deduction from your marks.

Task 1:

Install Outlook client

1.1 Create a test Gmail account or use your existing email account

1.2 Configure your Outlook client (Ref#1)



Using College account for this lab

1.3 Send/receive test emails

Test Email for Lab 8 Computer Forensics Inbox x



Shubham Pandya <shubham.pandya@dcmail.ca>

to me ▾

TEST TEST!

Great it is working 🥳



1.4 Viewing E-mail headers

Original Message

Message ID	<YT1PR01MB89223EAA321E4A190E28A04C9D51A@YT1PR01MB8922.CANPRD01.PROD.OUTLOOK.COM>
Created at:	Thu, Jul 17, 2025 at 6:14 PM (Delivered after 4 seconds)
From:	Shubham Pandya <shubham.pandya@dcmail.ca>
To:	"spandya.pshubham@gmail.com" <spandya.pshubham@gmail.com>
Subject:	Test Email for Lab 8 Computer Forensics
SPF:	PASS with IP 2a01:111:f403:c103:0:0:0:3 Learn more
DKIM:	'PASS' with domain dcmail.ca Learn more
DMARC:	'PASS' Learn more

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Text book page 460, 461 and 462.

Task-2:

Using Hex editor to carve email messages

2.1 Text book page 476, 477 and 478.

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00071510 3A 35 33 20 2D 30 30 30 0A 67 6F 6F 77 79 3A :53 -0000.goowy:
00071520 20 69 64 3A 20 3A 20 35 32 30 30 35 31 0A 46 72 id: : 520051 Fr
00071530 6F 6D 3A 20 74 65 72 72 79 73 61 64 6C 65 72 20 sm: terrysadler
00071540 3C 74 65 72 72 79 73 61 64 6C 65 72 40 67 6F 6F tterrysadler@goo
00071550 77 79 2E 63 6F 6D 3E 0A 52 65 70 6C 79 2D 54 6F y.com>.Reply-To
00071560 3A 20 74 65 72 72 79 73 61 64 6C 65 72 20 3C 74 : terrysadler <t
00071570 65 72 72 79 73 61 64 6C 65 72 40 67 6F 6F 77 79 rrysadler@goo
00071580 2E 63 6F 6D 3E 0A 54 6F 3A 20 6D 61 72 74 68 61 .com>.To: martha
00071590 64 64 61 78 40 73 75 70 65 72 69 6F 72 62 69 63 .dax@superiorbic
000715A0 79 63 6C 65 73 2E 62 69 7A 0A 44 61 74 65 3A 20 ycles.biz.Date:
000715B0 53 61 74 2C 20 31 37 20 46 65 62 20 32 30 30 37 Sat, 17 Feb 2007
000715C0 20 32 31 3A 31 35 3A 34 34 20 47 4D 54 0A 4D 65 21:15:44 GMT.Me
000715D0 73 73 61 67 65 2D 49 44 3A 20 3C 32 61 66 30 33 ssage-ID: <2af03
000715E0 31 35 38 34 62 35 63 34 36 30 65 39 35 62 33 36 1584b5c460e95b36
000715F0 64 64 64 36 37 31 39 35 32 39 66 40 77 65 62 73 ddd6719529f@weba
00071600 65 72 76 65 72 30 30 32 3E 0A 53 75 62 6A 65 63 rver002>.Subjct
00071610 74 3A 20 49 6E 76 65 73 74 6F 72 73 0A 4D 49 4D t: Investors.MIM
00071620 45 2D 56 65 72 73 69 6F 6E 3A 20 31 2E 30 0A 58 E-Version: 1.0.X
00071630 2D 4D 61 69 6C 65 72 3A 20 67 6F 6F 77 79 20 6D -Mailer: goowy m
00071640 61 69 6C 20 2D 20 68 74 74 70 3A 2F 77 77 77 77 ail - http://www
00071650 2E 67 6F 6F 77 79 2E 63 6F 6D 0A 50 72 69 6F 72 .goowy.com.Prior
00071660 69 74 79 3A 20 4E 6F 72 6D 61 6C 0A 58 2D 50 72 ity: Normal.X-Pr
00071670 69 6F 72 69 74 79 3A 20 33 0A 43 6F 6E 74 65 6E iority: 3.Conten
00071680 74 2D 54 79 70 65 3A 20 6D 75 6C 74 69 70 61 72 t-Type: multipar
00071690 74 2F 61 6C 74 65 72 6E 61 74 69 76 65 3B 20 62 t/alternative: b
000716A0 6F 75 6E 64 61 72 79 3D 22 2D 2D 2D 2D 3D 5F 45 oundary="----- E
000716B0 44 4E 50 5F 30 30 30 3F 36 61 63 64 37 34 35 DNP_0000_6acd745
000716C0 98 2D 64 65 63 62 2D 34 65 66 36 2D 62 63 38 63 8-daeb-4ef6-bc8c
000716D0 6D 64 34 37 38 38 65 30 39 30 31 39 62 22 0A 58 -d478a2a9019bX

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Note: Your Winhex editor's demo license may be expired, you may use other hex editors (Ex: HxD)

Task-3:

Examining Enron employee's email

3.1 Text book page 488 (Hands-On project 11-1)

Source Name	S	C	O	E-Mail From	E-Mail To	Subject
eric_saibl_000_1_1.pst				Saibl\u001a <Saibl\u001a>	muschar, Daniel	RE: request
eric_saibl_000_1_1.pst				Saibl\u001a <Saibl\u001a>	esaibl@houston.rr.com	Cinergy
eric_saibl_000_1_1.pst				Saibl\u001a <Saibl\u001a>	imat, Rika	RE: Virus Please kill
eric_saibl_000_1_1.pst				Saibl\u001a <Saibl\u001a>	muschar, Daniel	RE: request
eric_saibl_000_1_1.pst				Saibl\u001a <Saibl\u001a>	akhan@khanventures.com	eBay Auction
eric_saibl_000_1_1.pst				Saibl\u001a <Saibl\u001a>	forney, John M.	
eric_saibl_000_1_1.pst				Saibl\u001a <Saibl\u001a>	forney, John M.	eBay
eric_saibl_000_1_1.pst				Saibl\u001a <Saibl\u001a>	schlavone, Paul	
eric_saibl_000_1_1.pst				Saibl\u001a <Saibl\u001a>	gilbert-smith, Doug	

(There were errors in “Aid4Mail6” software so I instead used “Autopsy”)

A1	B1	C1	D1	E1	F1	G1	H1	I1	J1
Source Name									
Source Name	E-Mail From	E-Mail To	Subject	Date Recd Message (Plaintext)	Message ID Path	Thread ID	Data Source Has A		
eric_sabli_0001.1.pst	Sabli <Sabli>	Muschar; Daniel	RE: request	2001-12-2 Hey there,	2097188	/Top of Pe64cc23a-	LogicalFileSet1		
eric_sabli_0001.1.pst	Sabli <Sabli>	esabli@houston.rr.com	Cinergy	2001-12-1 Cinergy	2097220	/Top of Pe35e28c5d-	LogicalFileSet1		
eric_sabli_0001.1.pst	Sabli <Sabli>	imal; Rika	RE: Virus Please kill	2001-12-1 Just like any other day,	2097252	/Top of Pe15ea1e45-	LogicalFileSet1		
eric_sabli_0001.1.pst	Sabli <Sabli>	Muschar; Daniel	RE: request	2001-12-3 got it, thanks.	2097284	/Top of Pe64dc23a-	LogicalFileSet1		
eric_sabli_0001.1.pst	Sabli <Sabli>	akhanj@khanventures.com	eBay Auction	2002-01-1 Hey I sent this guy a food pr	2097316	/Top of Pe9425ea56-	LogicalFileSet1		
eric_sabli_0001.1.pst	Sabli <Sabli>	forney; John M.		2002-01-1 http://messages.yahoo.c	2097348	/Top of Peae81631a-	LogicalFileSet1		
eric_sabli_0001.1.pst	Sabli <Sabli>	forney; John M.	eBay	2002-01-1 http://cgi.ebay.com/aw-	2097380	/Top of Pe fad2aaec-	LogicalFileSet1		
eric_sabli_0001.1.pst	Sabli <Sabli>	schiavone; Paul		2002-01-1 http://cgi.ebay.com/aw-	2097412	/Top of Pe ae81631a-	LogicalFileSet1		
eric_sabli_0001.1.pst	Sabli <Sabli>	gilbert-smith; Doug		2002-01-1 http://cgi.ebay.com/aw-	2097444	/Top of Pe ae81631a-	LogicalFileSet1		
eric_sabli_0001.1.pst	Sabli <Sabli>	pjm1967@yahoo.com	FW: Question for seller -- Item #1064413482	2002-01-1 My co-worker here said	2097476	/Top of Pe e8beee06-	LogicalFileSet1		
eric_sabli_0001.1.pst	Sabli <Sabli>	meharban@duke-energy.com	FW: Question for seller -- Item #1061456076	2002-01-2 Hey there, I got a call from	2097508	/Top of Pe5ade0fa-	LogicalFileSet1		
eric_sabli_0001.1.pst	Sabli <Sabli>	esabli@houston.rr.com	FW: Message from Ken Lay	2002-01-2 -----Original Message-----	2097540	/Top of Pe d30aba5-	LogicalFileSet1		
eric_sabli_0001.1.pst	Sabli <Sabli>	cliffang@pacific.net.ph	RE: Hi!	2002-01-2 Hey Dude,	2097572	/Top of Pe64d1ce68-	LogicalFileSet1		
eric_sabli_0001.1.pst	Sabli <Sabli>	diamond; Daniel		2002-01-3 -----Original Message-----	2097604	/Top of Pe2647fc4d-	LogicalFileSet1		
eric_sabli_0001.1.pst	Sabli <Sabli>	carson; Mike; kanis; Jason	Load Mix	2002-02-0 Here's the load mix for Sou	2097636	/Top of Pe5afdd1a7-	LogicalFileYes		
eric_sabli_0001.1.pst	Sabli <Sabli>	esabli@houston.rr.com	Auto Ins	2002-02-0 6 mnnth 575 15-20%	2097668	/Top of Pe5ced7a80-	LogicalFileSet1		
eric_sabli_0001.1.pst	Sabli <Sabli>	esabli@houston.rr.com	FW: Regional Research Projects	2002-02-0 -----Original Message-----	2097700	/Top of Pe faada9e4-	LogicalFileYes		
eric_sabli_0001.1.pst	Sabli <Sabli>	valdes; Maria	RE: Regional Research Projects	2002-02-0 Here is my homework	2097732	/Top of Pe fecf197-	LogicalFileYes		
eric_sabli_0001.1.pst	Lance Cunningham<Lance Cunn Sel Yaman; martin.lins		Power World drawing and contingency files	2001-04-3 Sevil and others,	2097764	/Top of PeSc30M44-	LogicalFileYes		
eric_sabli_0001.1.pst	Day Smith L.Jay<ENR Miller Jeffrey; Gilbert-Smith Doug		ERCOT RESPONSIBILITY TRANSFERS AND DYNAMIC SCHEDULES	2001-06-0 When: Friday, July 06,	2097796	/Top of P eaSe4ed46-	LogicalFileSet1		
eric_sabli_0001.1.pst	Will Lloyd<Lloyd.Will@ENRON.C> Sabli Eric		Erctot Certificate	2001-05-3 Attached is you digital	2097828	/Top of Pe 276257-	LogicalFileYes		
eric_sabli_0001.1.pst	Day Smith L.Jay<ENR Miller Jeffrey; Gilbert-Smith Doug		ERCOT RESPONSIBILITY TRANSFERS AND DYNAMIC SCHEDULES	2001-06-0 When: Wednesday, June	2097860	/Top of P eaSe4ed46-	LogicalFileSet1		
eric_sabli_0001.1.pst	Forney<Forney>	erriego; Joe; salbie Eric	FRONTIER NUMBERS FOR JERRY WATSON AT FRONTERA/TE	2001-07-1 Please use the following:	2097892	/Top of Pe 0a4eb4d8-	LogicalFileSet1		
eric_sabli_0001.1.pst	Forney<Forney>	dean; Clint; olbine Jr.; Steve	Alterna Backup person	2001-07-2 Stephanie McHargue will	2097924	/Top of Pc f17a210-	LogicalFileSet1		
eric_sabli_0001.1.pst	Day Smith L.Jay<ENR Capasso Joe; Errigo Joe; mcleareth		How to Copy ERCOT Weekend Schedules	2001-08-1 Joe E.	2097956	/Top of Pe5450fbfe-	LogicalFileYes		
eric_sabli_0001.1.pst	Day Smith L.Jay<ENR Capasso Joe; Errigo Joe; mcleareth		FW: RFIIANT-HLP real time FRCOT Schedules	2001-08-2 -----Inline attachment	2097988	/Top of Pc c030775-	LogicalFileSet1		

Ans: Just from the email metadata much information can be retrieved such as sender, receiver, timestamp, subject of email, message ID, and Authentication details. This information can be helpful in linking email with other evidence that are physical or like email i.e. logical or in data. This metadata can be used to determine if the email is spoofed or not.

5. Why are network router logs important during an e-mail investigation?

Ans: Network router logs are important because they help provide various information like IP addresses, unauthorized connections and if any incident occurs it is helpful to reset timeline back to its original and know for how long the system was offline or out of service. It is also helpful in verifying of various evidence such as emails sent from the organization's mail and other activities during active hours or during incident.

References

- 1) Nucleus Technologies. (n.d.). How to configure Gmail in MS Outlook?
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<https://core.ac.uk/download/pdf/56362354.pdf>
- 4) McCann, C. (2005). IDS file forensics. SANS Institute.
https://digital-forensics.sans.org/community/papers/gcfa/ids-file-forensics_9991